A Raindrop's Journey: Approximating Precipitation Efficiency

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BACKGROUND

- Precipitation Efficiency (η) = The amount of a raindrop that makes it to the surface
- Importance of η : has implications for relative humidity, or how far the air is from saturation, at the Earth's surface[1] and the water cycle[2], meaning it has a significant impact on ecological processes[1]
- η is poorly understood within the climate system[3], so we need better constraints to improve idealized weather modeling.
- Raindrop evaporation rate [4]:

$$r\frac{dr}{dt} = -\frac{D_v}{\rho_w R_v}\left(\frac{e_d^*}{T_d} - H\frac{e_a^*}{T_a}\right) f_v$$

- Falling raindrop ventilation factor [4]:

$$f_v = a_v + b_v Sc^{1/3} Re^{1/2}$$

- The Clausius-Clapeyron expression for saturation specific humidity [4]:

$$q_s^*(T,p) = \frac{R_d}{R_v}\frac{e_o^*}{p} e^{-\frac{L_v}{R_v}\left(\frac{T_o - T}{T T_o}\right)}$$

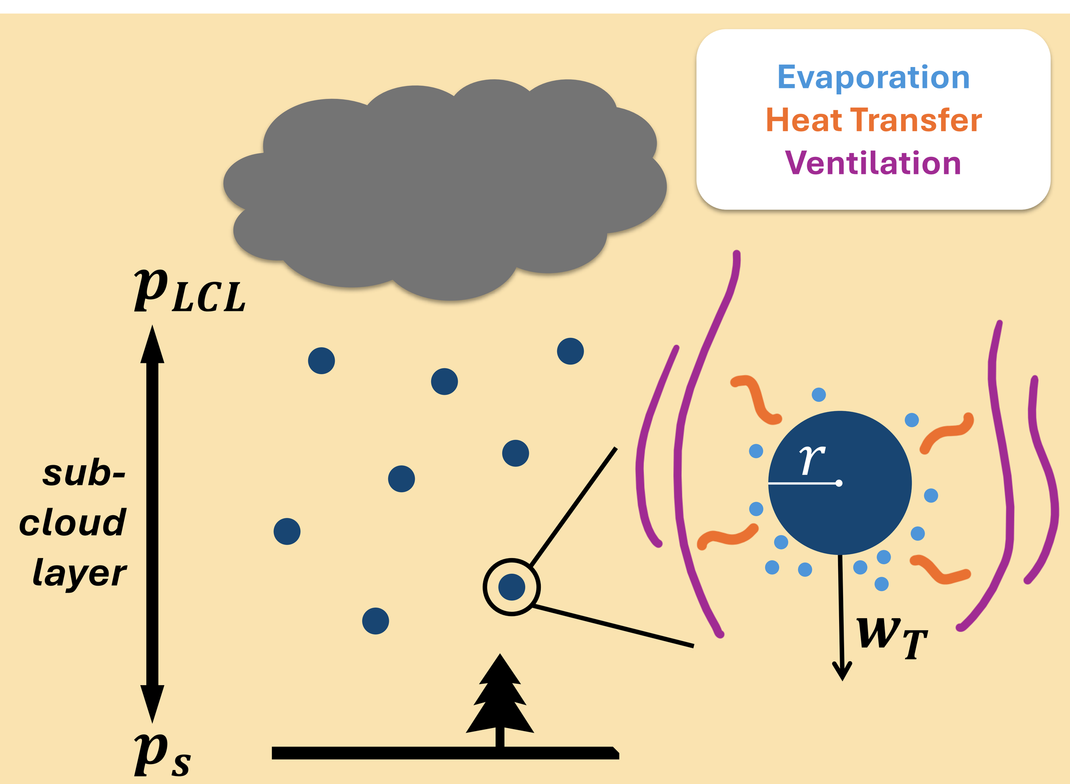

Figure 1: Diagram of precipitation efficiency. The evaporation of the raindrop as it falls through the subcloud layer at terminal velocity w_T is driven by ventilation due to the flow of air past the drop and the heat transfer away from the drop due to the phase change.

GOAL & ASSUMPTIONS

- We aim to derive a simple analytic approximation of precipitation efficiency for a single raindrop as a function of surface relative humidity (H), surface air temperature (T_s), and initial radius of the raindrop (r_{LCL}).
- We assume a dry adiabat through the subcloud layer and a population of raindrops starting with the same radius.

THEORY & ASSESSMENT

The Process

Account for Ventilation
Power Law Simplification of Ventilation Factor

↓

Change Time Dependence to Pressure Dependence
Hydrostatic Balance, Power Law Simplification of Terminal Velocity[5]

↓

Integrate over the Subcloud Layer
Surface Value Approximations

↓

Put the Expression in Terms of Relative Humidity
Power Law Simplification of the Clausius-Clapeyron Equation

Our Approximation

$$[1]\ \eta \approx \left(\frac{r_s}{r_{LCL}}\right)^3 \approx [1 - KS]^{3/(2+b-c)}$$

K accounts for properties of the raindrop.

S accounts for properties of the environment.

$$[2]\ K = b_v Re^{1/2} D_{v_o} \frac{(3+b)}{\sqrt{2}\, w_{T_{LCL}}\, r_{LCL}^2\, \rho_w}\left(\frac{1}{Sc}\right)^{1/3}$$

$$[3]\ S = \frac{q_s^* p_s}{g\left(\frac{19}{14}+\gamma\right)}\chi(T_{ref}) H^{-5/14\gamma}\left(1 - H\left(\left(1+\frac{14}{19}\gamma\right) - \frac{14}{19}\gamma H^{19/14\gamma}\right)\right)$$

$$[4]\ \chi(T_{ref}) = \left(1 + \frac{L_v\, q^*(T_{ref}, p_s)}{c_p T_{ref}}\left(\frac{L_v}{R_v T_s} - 1\right)\right)^{-1}$$

$$[5]\ c = \left(b_v Sc^{1/3} Re_{LCL}^{1/2}\left(\frac{1+b}{2}\right)\right) * \left(a_v + b_v Sc^{1/3} Re_{LCL}^{1/2}\right)^{-1}$$

[1] Full approximation of precipitation efficiency
[2] Effective Evaporative Cross-Section: The area of the raindrop which is undergoing evaporation, adjusted to account for ventilation and fall time of the raindrop (units of m^2/kg)
[3] Effective Saturation Deficit: The thermodynamic potential for evaporation, based on surface air temperature and relative humidity (units of kg/m^2)
[4] Decrease in saturation vapor pressure of the drop due to cooling at the drop surface from evaporation
[5] Portion of the exponent which captures the full ventilation factor term

What does this tell us?

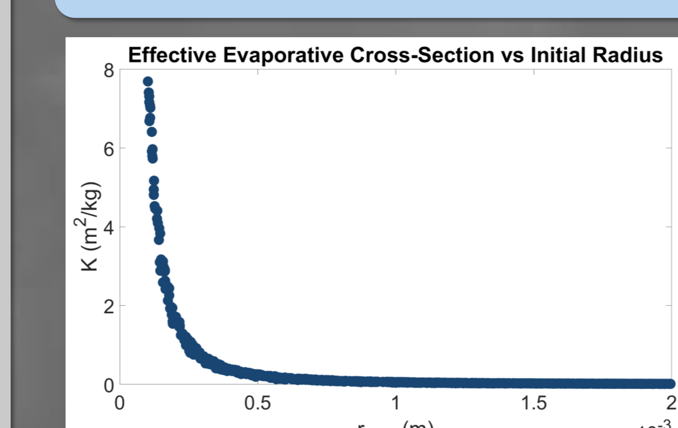

Figure 2: K is large for small drops and approaches zero for large drops.

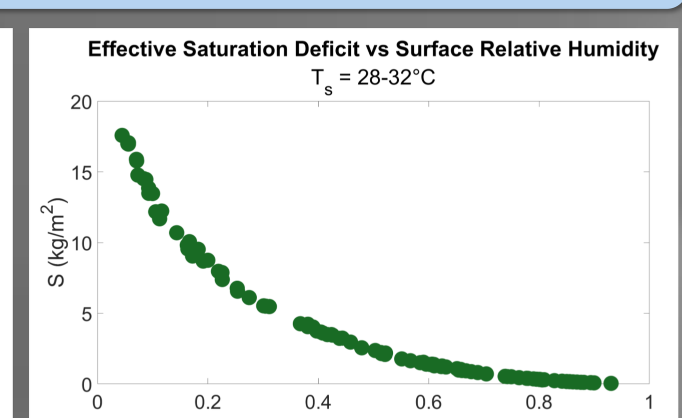

Figure 3: S decreases as relative humidity approaches 1.

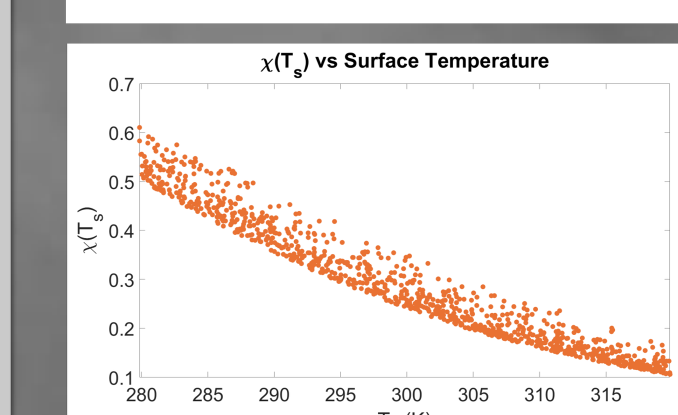

Figure 4: The impact of heat transfer contributes to higher η for higher T_s.

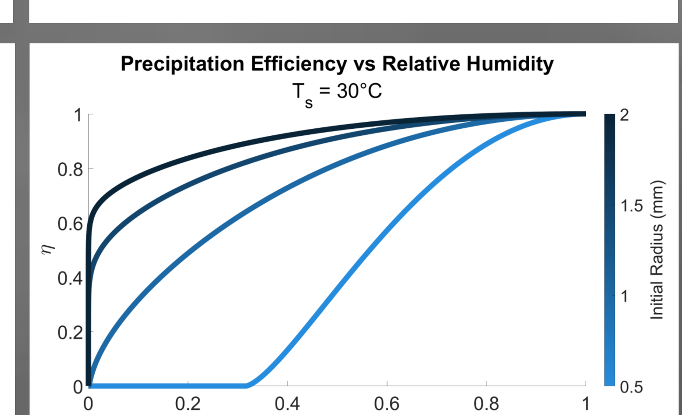

Figure 5: η is lower when relative humidity is lower. The effect is larger for smaller raindrops.

How well did we do?

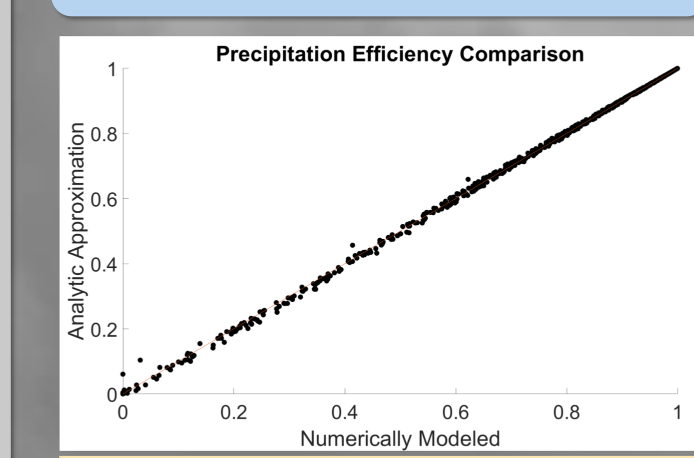

Figure 6

Statistical Accuracy

- *R^2 for 1:1 Line* = 0.9997

Our approximation captures the patterns of the data quite well.

- *Absolute Mean Error* = 0.0034

Our approximation's predictions are very close to the numerical predictions.

DISCUSSION

- Faster falling drops, though f_v is greater, have a greater η because they spend less time in the subcloud layer.
- The high R-Squared value for the 1:1 line and low Absolute Mean Error value produced when comparing the results of the approximated precipitation efficiency to the numerically-modeled precipitation efficiency show that the ventilation and Clausius-Clapeyron simplifications we made are performing well.
- Overall, our precipitation efficiency approximation seems to be working well as a way to quantify how much of an impact surface relative humidity, surface air temperature, and initial raindrop size have on precipitation efficiency.

FUTURE DIRECTIONS

- Determine the best reference temperature for $\chi(T_{ref})$
- Integrate $\chi(T)$ over the subcloud layer
- Account for raindrop shape in the terminal velocity expression rather than assuming spherical drops
- Integrate over a range of initial raindrop radii
- Compare the approximation with observations of raindrop evaporation to further validate our results

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[7] The background photo is courtesy of Pixabay.